STAT/MA 41600
In-Class Problem Set #27: October 21, 2026
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Problem Set 27 Answers

1a. For $0 < x < 8$, we learned in Problem Set 26 that $f_X(x) = (1/32)(8-x)$. It follows that $f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{1/20}{(1/32)(8-x)} = \frac{8}{(5)(8-x)}$ for $0 < y < 5 - 5x/8$, and otherwise $f_{Y|X}(y \mid x) = 0$.

1b. For $0 < y < 5$, again from Problem Set 26, we know that $f_Y(y) = (2/25)(5-y)$. It follows that $f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{1/20}{(2/25)(5-y)} = \frac{5}{(8)(5-y)}$ for $0 < x < 8 - 8y/5$, and otherwise $f_{X|Y}(x \mid y) = 0$.

2a. We have $f_{Y|X}(y \mid 1) = \frac{f_{X,Y}(1,y)}{f_X(1)} = \frac{8}{(5)(8-1)} = \frac{8}{35}$ for $0 < y < 35/8$, and otherwise $f_{Y|X}(y \mid 1) = 0$. It follows that $P(Y \leq 2 \mid X = 1) = \int_0^2 f_{Y|X}(y \mid 1)\, dy = \int_0^2 8/35\, dy = (8/35)(2) = 16/35$.

2b. We have $P(Y \leq 2 \mid X \leq 1) = \frac{P(Y \leq 2 \ \& \ X \leq 1)}{P(X \leq 1)}$. We can either calculate the ratio of areas $\frac{2}{75/16} = \frac{32}{75}$, or we can calculate: $\frac{P(Y \leq 2 \ \& \ X \leq 1)}{P(X \leq 1)} = \frac{\int_0^1 \int_0^2 1/20\, dy\, dx}{\int_0^1 \int_0^{5-5x/8} 1/20\, dy\, dx} = \frac{1/10}{15/64} = \frac{32}{75}$.

3a. For $0 < x < 5$, we learned in Problem Set 26 that $f_X(x) = 4/15 - 2x/75$. It follows that $f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{(4/1875)(25-xy)}{4/15-2x/75}$ for $0 < y < 5$, and otherwise $f_{Y|X}(y \mid x) = 0$.

3b. We have $f_{Y|X}(y \mid 2) = \frac{f_{X,Y}(2,y)}{f_X(2)} = \frac{(4/1875)(25-2y)}{4/15-4/75} = (25-2y)/100$ for $0 < y < 5$, and otherwise $f_{Y|X}(y \mid 2) = 0$. It follows that $P(Y \leq 1 \mid X = 2) = \int_0^1 f_{Y|X}(y \mid 2)\, dy = \int_0^1 (25-2y)/100\, dy = 6/25$.

3c. We have $P(Y \leq 1 \mid X \leq 2) = \frac{P(Y \leq 1 \& X \leq 2)}{P(X \leq 2)} = \frac{\int_0^2 \int_0^1 \frac{4}{1875}(25-xy)\, dy\, dx}{\int_0^2 \int_0^5 \frac{4}{1875}(25-xy)\, dy\, dx} = \frac{\int_0^2 \int_0^1 (25-xy)\, dy\, dx}{\int_0^2 \int_0^5 (25-xy)\, dy\, dx} = \frac{49}{225}$.

4a. For $x > 0$, we learned in Problem Set 25 that $f_X(x) = 9e^{-9x}$. It follows that $f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{63e^{-2x-7y}}{9e^{-9x}} = 7e^{-7(y-x)}$ for $y > x > 0$, and otherwise $f_{Y|X}(y \mid x) = 0$.

4b. For $y > 0$, we learned in Problem Set 26 that $f_Y(y) = \frac{63}{2}(e^{-7y} - e^{-9y})$. It follows that $f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{63e^{-2x-7y}}{(63/2)(e^{-7y}-e^{-9y})} = \frac{2e^{-2x}}{1-e^{-2y}}$ for $y > x > 0$, and otherwise $f_{X|Y}(x \mid y) = 0$.

5a. We have $f_{X|Y}(x \mid 5) = \frac{f_{X,Y}(x,5)}{f_Y(5)} = \frac{63e^{-2x-(7)(5)}}{\frac{63}{2}(e^{-(7)(5)}-e^{-(9)(5)})} = \frac{2e^{-2x}}{1-e^{-10}}$ for $5 > x > 0$, and otherwise $f_{X|Y}(x \mid 5) = 0$. It follows that $P(X \leq 3 \mid Y = 5) = \int_0^3 f_{X|Y}(x \mid 5)\, dx = \int_0^3 \frac{2e^{-2x}}{1-e^{-10}}\, dx = (1-e^{-6})/(1-e^{-10})$.

5b. We have $P(X \leq 3 \,|\, Y \leq 5) = \frac{P(X \leq 3 \ \& \ Y \leq 5)}{P(Y \leq 5)} = \frac{\int_0^3 \int_x^5 63e^{-2x-7y}\, dy\, dx}{\int_0^5 \int_x^5 63e^{-2x-7y}\, dy\, dx} = \frac{1-(9/2)e^{-35}+(9/2)e^{-41}-e^{-27}}{1-(9/2)e^{-35}+(7/2)e^{-45}}$.